

Rahul SUBRAMANIAN GIRIJA

PhD - Materials Science

Materials Scientist with a PhD and specialized expertise in advanced materials for aerospace applications. Proven track record in R&D, process optimization and materials characterization within a tier 1 aerospace supplier. Passionate about driving innovation and solving complex materials challenges to enhance product performance and reliability. Seeking to leverage my technical expertise and industry experience in a dynamic R&D or engineering role.

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SKILLS

Scientific Research | R&D | Materials Science | Materials Characterization | Market Research | Metal Additive Manufacturing | Experimental investigation | Process Optimization | Project Management | Numerical Simulation | Fatigue of metals | Corrosion | Hydrogen Embrittlement | Heat Exchangers

WORK EXPERIENCE

Mechanical and Physical Engineer
Capgemini Engineering

04/2024 - Present — *Toulouse, France*

Scientific Specialist for Materials and Special Processes team at Liebherr Aerospace Toulouse

- **Compact Aluminium heat exchangers for Next Single Aisle** - developed a solution for high quality brazing joints with 80µm fins avoiding clad diffusion in the joints during brazing through extensive experimental and microstructural analyses to attain specific microstructural characteristics that result in an optimal brazing joint, that led implementing changes to fin forming process at Liebherr.
- **Graphite for Seal Rings in valves** - Developed a comprehensive material characterization framework (ASTM-compliant) and supplier qualification protocol, enabling end-to-end material evaluation and vendor assessment for enterprise applications.
- **Inconel 718 for valve body redesign** - Led microstructural characterization (casting, forging, additive manufacturing) linking microstructure to mechanical properties post-heat treatment, enabling optimal material selection for service conditions at high temperatures.
- Successfully transitioned from academic research to corporate R&D, rapidly adapting to industry workflows, cross-functional collaboration, and applied problem-solving to deliver actionable material solutions.

PhD / Young Researcher
I2M, National Centre for Scientific Research (CNRS)

02/2020 - 08/2023 — *Bordeaux, France*

Academic Research in collaboration with Institut P', LASIE and MCAM, Australia.

- Evaluated **Hydrogen Embrittlement (HE)** on Laser – Powder Bed Fusion (L-PBF) **316L, 304L and 17-4PH stainless steels** under tensile and High Cycle Fatigue loading conditions.
- Developed **coupled fatigue-hydrogen** electrochemical testing apparatus. Worked with high pressure hydrogen gas mechanical testing machines. Carried out **extensive experimental campaigns** between 3 academic research labs.
- **Spearheaded procurement** of material powders and lab equipment by managing **internal/external communications,** coordinating with suppliers, and securing competitive quotations to support lab operations.
- Performed fabrication, machining and characterisation of materials followed by frequent **scientific communication** of the analysed results. Mentored various students for Masters projects and internships around the thematic.

LANGUAGES

English
Native or Bilingual Proficiency

French
Full Professional Proficiency

TECHNICAL KNOWLEDGE

Materials Characterization: Microscopy, Alicona, Keyence, SEM, TEM, XRD, EBSD, EDS, X-ray Tomography, AFM, TDS, GD-OES, Mechanical testing (tension, fatigue, (nano) hardness),

Numerical: OIM Analysis, ATEX, FEM, Zebulon, Meshlab, Gmsh AutoCAD, CATIA, Autodesk Inventor, Python, MATLAB, Materialise MAGICS, Avizo.

Scientific Skills: Scientific thinking, analysis and writing, Technical Presentation, Problem solving, Sofware : Origin Pro, Office 365, LaTeX.

Soft Skills: Listening, Team Work, Transparency, Being Empathetic.

EDUCATION

PhD - Materials Science
HESAM University - ENSAM - Institut de Mécanique et d'Ingénierie (I2M)

02/2020 - 02/2024 — *Bordeaux, France*

Master – Aeronautical Mechanics and Energetics (AME)
ISAE-ENSMA École Nationale Supérieure de Mécanique et d'Aérotechnique

09/2017 - 09/2019 — *Poitiers, France*

- **Specialisation : High Temperature Materials**

SCIENTIFIC PROJECTS

Master-2 Research Internship - Institut P' / UGITECH (04/2019 - 10/2019)
- **Hydrogen Embrittlement assessment of stainless steels at low temperatures.**

Masters Research Project with Safran Aircraft Engines (09/2018 - 02/2019)
- **Effect of Over-aging on the creep behaviour of cast Inconel 718 alloy**.

Master-1 Research Internship - Institut P' / Project ANR ImPEKKable (04/2018 - 10/2018)
- **Nanoindentation Creep Behaviour of Epoxy, Aged-Epoxy and PEKK polymer at high temperatures.**

Independant Student Team Project for Student Aerospace Challenge'18 (11/2017 - 07/2018)
- **Study of heat transfer modes and materials for a cryogenic liquid oxygen tank to reduce excessive vaporisation and improve design.**
- **AWARDED ARIANE GRAND PRIZE FOR THE BEST PROJECT.**

Bachelor Final Year Internship (11/2016 - 04/2017)
- **Investigation of Acoustic Emission Behaviour of glass-fibre composite laminates subject to a small-scale impact.**

PUBLICATIONS

Conference Proceedings

Assessment of the influence of hydrogen on the fatigue life of selective laser melted 316L austenetic stainless steels

Subramanian Girija et al.

Nicolas Saintier, Gilbert Henaff, Isabelle Aubert, Mohamed El May, Abdelali Oudriss, Xavier Feaugas

2022

International Conference on Metals and Hydrogen - Steelyhydrogen

CERTIFICATES

Emerging Leaders Program - Grade B (06/2025 - 07/2025)

Leadership development program offered by Capgemini in colloboration with Harvard Business Review

Sustainability for Products and Systems Engineers (08/2025 - 09/2025)

Beginner course on Sustainability, Systems Thinking, Eco-Design, Life-Cycle Analysis, Sustainable Energy Sources.